

Problem Set 5

Ejercicio #1

$$f(x) = \frac{1}{2} \left[(x_1)^{1/3} + (x_2)^{1/3} \right]^2$$

1)
$$f(tx) = \frac{1}{2} \left[t^{1/3} (x_1^{1/3} + x_2^{1/3}) \right]^2, \quad t > 1$$

$$f(tx) = t^{2/3} f(x).$$

$\Rightarrow f(tx) > t f(x)$ Rendimientos decrecientes

2).
$$\min_{x_1, x_2} w_1 x_1 + w_2 x_2 \quad \text{s.t.} \quad f(x) \geq \bar{y}$$

$$\mathcal{L}(x_1, x_2, \lambda) = w_1 x_1 + w_2 x_2 - \lambda (x_1^{1/3} + x_2^{1/3} - \sqrt{2\bar{y}})$$

C.P.O

$$\begin{aligned} [x_1]: \quad w_1 - \lambda \frac{1}{3} x_1^{-2/3} &= 0 \\ [x_2]: \quad w_2 - \lambda \frac{1}{3} x_2^{-2/3} &= 0 \end{aligned} \quad \Rightarrow \quad \left(\frac{w_1}{w_2} \right)^{3/2} = \frac{x_2}{x_1}$$

$$[\lambda]: \quad x_1^{1/3} + x_2^{1/3} = \sqrt{2\bar{y}}$$

$$x_1^{1/3} + \left(x_1 \left(\frac{\omega_1}{\omega_2} \right)^{3/2} \right)^{1/3} = \sqrt{2\bar{y}}$$

$$x_1^{1/3} \left(1 + \left(\frac{\omega_1}{\omega_2} \right)^{1/2} \right) = \sqrt{2\bar{y}}$$

$$x_1^C = \left(\frac{\sqrt{2\bar{y}}}{1 + \left(\frac{\omega_1}{\omega_2} \right)^{1/2}} \right)^3 = \left(\frac{\sqrt{2\bar{y}} \omega_2}{\omega_2^{1/2} + \omega_1^{1/2}} \right)^3 //$$

Por simetría

$$x_2^C = \left(\frac{\sqrt{2\bar{y}}}{1 + \left(\frac{\omega_2}{\omega_1} \right)^{1/2}} \right)^3 = \left(\frac{\sqrt{2\bar{y}} \omega_1}{\omega_2^{1/2} + \omega_1^{1/2}} \right)^3 //$$

⇒ Función de costo:

$$C(\omega, \bar{y}) = \omega_1 x_1^C + \omega_2 x_2^C$$

$$= \frac{(2\bar{y})^{3/2} (\omega_1 \cdot \omega_2^{3/2} + \omega_2 \cdot \omega_1^{3/2})}{(\omega_1^{1/2} + \omega_2^{1/2})^3}$$

$$C(\omega, \bar{y}) = \frac{\omega_1 \omega_2}{(\omega_1^{1/2} + \omega_2^{1/2})^2} \cdot (2\bar{y})^{3/2} //$$

$$3) \max_y P \cdot y - C(w, y)$$

$$\max_y P \cdot y - \frac{w_1 w_2}{\underbrace{(w_1^{1/2} + w_2^{1/2})^2}_A} (2y)^{3/2}$$

C.P.O.

$$[y]: P - A \frac{3}{2} \cdot 2^{3/2} y^{1/2} = 0$$

$$\frac{2}{3} \cdot \frac{P}{A} \cdot \frac{1}{2^{3/2}} = y^{1/2}$$

$$y = \frac{4}{9} \cdot \frac{1}{8} \left(\frac{P}{A} \right)^2$$

$$y = \frac{1}{18} \left(\frac{P}{A} \right)^2 //$$

Demand incondicional.

$$X(P, w) = x^c(w, y)$$

$$X_1(P, w) = \left(\frac{\sqrt{2 w_2} Y(P, w)^3}{w_1^{1/2} + w_2^{1/2}} \right)^3$$

$$= \left(\frac{\left(2 w_2 \frac{1}{18} P^2 \frac{(w_1^{1/2} + w_2^{1/2})^4}{(w_1 w_2)^2} \right)^{1/2}}{w_1^{1/2} + w_2^{1/2}} \right)^3$$

$$= \left(\frac{1}{9} \frac{\sqrt{2\omega_2} P(\omega_1^{1/2} + \omega_2^{1/2})}{\omega_1 \omega_2} \right)_{//}^3$$

$$X_2(P, \omega) = \left(\frac{1}{9} \frac{\sqrt{2\omega_1} P(\omega_1^{1/2} + \omega_2^{1/2})}{\omega_1 \omega_2} \right)_{//}^3$$

$$4) \pi(P, \omega) = P \cdot y(P, \omega) - C(\omega, y(P, \omega))$$

$$= P \cdot y(P, \omega) - \frac{\omega_1 \omega_2}{(\omega_1^{1/2} + \omega_2^{1/2})^2} (2y(P, \omega))^{3/2}$$

$$= y(P, \omega) \left(P - 2^{3/2} \frac{\omega_1 \omega_2}{(\omega_1^{1/2} + \omega_2^{1/2})^2} y(P, \omega)^{1/2} \right)$$

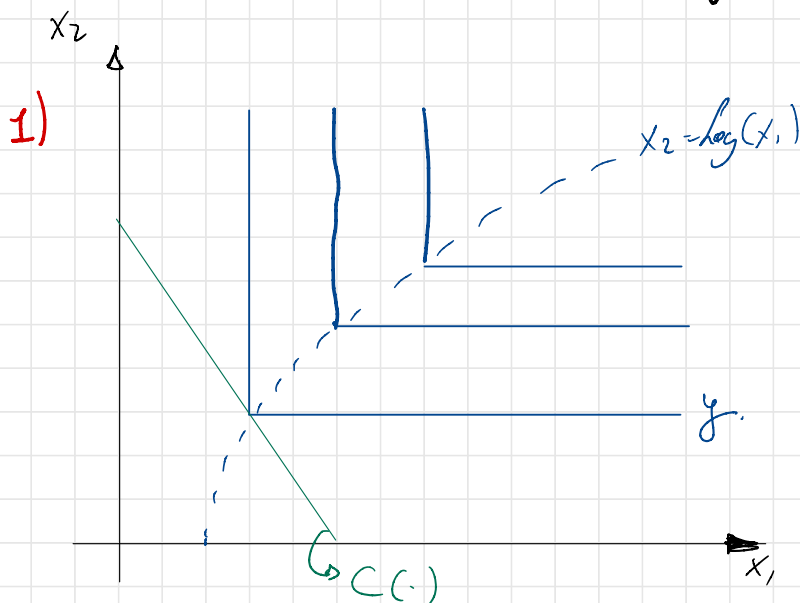
$$= \frac{1}{18} \frac{P^2 (\omega_1^{1/2} + \omega_2^{1/2})^4}{(\omega_1 \omega_2)^2} \left(P - 2^{3/2} \frac{\omega_1 \omega_2}{(\omega_1^{1/2} + \omega_2^{1/2})^2} \frac{1}{18^{1/2}} \frac{P (\omega_1^{1/2} + \omega_2^{1/2})^2}{\omega_1 \omega_2} \right)$$

$$= \frac{1}{18} \frac{P^3 (\omega_1^{1/2} + \omega_2^{1/2})^4}{(\omega_1 \omega_2)^2} \left(1 - \left(\frac{4}{18} \right)^{1/2} \right)$$

$$= \frac{1}{54} \frac{P^3 (\omega_1^{1/2} + \omega_2^{1/2})^4}{(\omega_1 \omega_2)^2}$$

Ejercicio 2.

$$f(x) = \min \{ \log(x_1), x_2 \}$$



2) La función es sobre inputs complementarios.

$\Rightarrow$ la solución está en el ángulo recto de las isocuantas.

$$\Rightarrow \log(x_1) = \bar{y} ; \quad x_2 = \bar{y}$$

$$x_1^c(w, \bar{y}) = e^{\bar{y}}, \quad x_2^c(w, \bar{y}) = \bar{y}$$

$$C(w, \bar{y}) = w_1 e^{\bar{y}} + w_2 \bar{y}$$

$$3) \max_y P \cdot y - C(w, y)$$

$$\max_y P \cdot y - w_1 e^y - w_2 y$$

C.P.O

$$[y]: P - w_1 e^y - w_2 = 0$$

$$\frac{P - w_2}{w_1} = e^y$$

$$\ln \left(\frac{P - w_2}{w_1} \right) = y$$

$$\text{For, } y > 0 \Rightarrow \frac{P - w_2}{w_1} > 1 \Rightarrow P - w_2 > w_1$$

$$\text{Si } P - w_2 \leq w_1 \Rightarrow y = 0.$$

$$\begin{cases} y = \ln \left(\frac{P - w_2}{w_1} \right) & \text{si } P - w_2 > w_1 \\ y = 0 & \text{si } P - w_2 \leq w_1 \end{cases}$$

$\Rightarrow$ Demanda incondicional.

$$X_i(P, w) = X_i^C(w, y(P, w))$$

$$X_1^C(P, w) = e^{\ln \left(\frac{P - w_2}{w_1} \right)} = \frac{P - w_2}{w_1} //$$

$$X_2^C(P, w) = \ln \left(\frac{P - w_2}{w_1} \right) //$$

$$4) \quad \pi(p, w) = p \cdot y(p, w) - w_1 e^{y(p, w)} - w_2 y(p, w)$$

$$= p \ln\left(\frac{p-w_2}{w_1}\right) - (p-w_2) - w_2 \ln\left(\frac{p-w_2}{w_1}\right)$$

$$= \ln\left(\frac{p-w_2}{w_1}\right) (p-w_2) - (p-w_2)$$

$$= (p-w_2) \left(\ln\left(\frac{p-w_2}{w_1}\right) - 1 \right)$$

$$\pi(p, w) = \begin{cases} (p-w_2) \left(\ln\left(\frac{p-w_2}{w_1}\right) - 1 \right) & \text{si } p-w_2 > w_1 \\ -w_1 & \text{si } p-w_2 \leq w_1 \end{cases}$$